PACKING $k^2 - \lceil\sqrt{32k}\,\rceil - 1$ UNIT SQUARES IN A SQUARE OF SIDE LESS THAN k

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ABSTRACT. Let $s(N)$ be the side of the smallest square containing N unit squares with disjoint interiors, and let $c^*(k)$ be the largest c with $s(k^2 - c) = k$. We prove $c^*(k) < 4\sqrt{2k} + 1$ for every integer $k \geq 6$, that is, $s(k^2 - \lceil\sqrt{32k}\,\rceil - 1) < k$. The construction is Ryu's L-shaped packing: an aligned grid and two bands of b tilted rows. Both ends of each band are filled with axis-parallel rows, and b is the odd integer nearest $\sqrt{2k}$. The odd width keeps the rounding loss of each filled end below $3/2$. This lowers the constant of the $O(\sqrt{k})$ bound valid for every $k \geq 6$ from 8 (Ryu) to $4\sqrt{2}$. The bound is the smallest explicit one we know of for $42 \leq k < 1.6 \cdot 10^7$, except at five values of k where Ryu has computer-assisted bounds. Exact rational certificates for $k \leq 600$ are checked by two independently written programs.

1. INTRODUCTION

For $x > 0$ let $N(x)$ be the largest number of unit squares that fit, with pairwise disjoint interiors and rotations allowed, in a square of side x. For an integer k, $N(k) = k^2$. Let

$$M(k) = \max\{N : s(N) < k\}, \qquad w(k) = k^2 - M(k), \qquad c^*(k) = \max\{c : s(k^2 - c) = k\}.$$

Then $s(k^2 - c) < k$ exactly when $M(k) \geq k^2 - c$, so $c^*(k) = w(k) - 1$.

Erdős and Graham [EG75] showed that the waste $x^2 - N(x)$ is $O(x^{7/11})$, which makes $w(k)$ sublinear. The best exponent is now $3/5$ [Bui25, McC26] (the earlier claim [CG20] contained an error [AB26]). For integer sides approached from below, Ryu [Ryu26] proved, among other things:

- $c^*(k) \leq 8\lceil\sqrt{k-4}\,\rceil - 1$ for every $k \geq 6$ (his Theorem 1.4), from an explicit L-shaped packing (his Theorem 1.3);
- $c^*(k) < 20.668\,k^{2/5} + 0.623$ for $k \geq 1.6 \cdot 10^7$ (Theorem 1.2), and bounds of order $k^{3/8}$ from $k \geq 2.48 \cdot 10^{12}$ (Theorem 1.1);
- computer-assisted bounds at five values of k between 10^5 and 10^8 (Theorem 1.6).

Lower bounds of order $\log k$, $k^{1/4}$ and $k^{1/3}$ for large k are also due to Ryu [Ryu26b]. Arslanov, Mustafin and Shangitbayev [AMS21] proved $c^*(k) \leq k - 1$ for $k \geq 12$. In an earlier note [Fan26a] we proved $c^*(k) \leq k - 1 - F(k)$ for $k \geq 15$, with $F(k) = 1 + \max\{p + 2q : 16p + 21q \leq k - 15\}$.

Theorem 1. *For every integer $k \geq 6$, $w(k) < 4\sqrt{2k} + 2$. Equivalently $c^*(k) < 4\sqrt{2k} + 1$, so $c^*(k) \leq \lceil\sqrt{32k}\,\rceil$, and $s(n) < k$ for every integer $n \leq k^2 - \lceil\sqrt{32k}\,\rceil - 1$.*

The packing is Ryu's L (his Sections 3–4), and we use his end-region framework (his Section 6): lifted band ends, the far end as the 180° image of the near one, and end fills assembled with the rows. What is new is the fill (axis-parallel rows at every integer level of each end region), the choice of an odd band width, and the estimate of Lemma 2. To keep the note self-contained, we prove everything we use.

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2. The band

Fix an integer $b \geq 3$ and a rational $t = \tan(\theta/2) > 1/b$. Write $c = \cos\theta = (1-t^2)/(1+t^2)$, $s = \sin\theta = 2t/(1+t^2)$, $T = s/c$, $\tau = c/s$, $\sec = 1/c$, and

$$W = bc + s \quad \text{(band width)}, \qquad h = bs + c \quad \text{(row height)}.$$

At $t_0 = 1/b$: $c_0 = (b^2-1)/(b^2+1)$, $s_0 = 2b/(b^2+1)$, $\tau_0 = (b^2-1)/(2b)$, $W_0 = b$, $h_0 = (3b^2-1)/(b^2+1) < 3$. Since $dW/d\theta = c - bs < 0$ for $\tan\theta > 1/b$, and $\theta_0 = 2\arctan(1/b) > \arctan(1/b)$, we have $W < b$ for every $t > 1/b$.

Use band coordinates $X \in [0, W]$ (across) and $Y \in [0, L]$ (along). With (u, v) the coordinates along (c, s) and $(-s, c)$, the rectangle $P = \{0 \leq u \leq b,\ 0 \leq v \leq 1\}$ is the union of the b unit squares $\{i \leq u \leq i+1\}$. *Row* j is P translated so that its corner $u = v = 0$ is at $(s,\ y_0 + j\sec)$, for $0 \leq j < R$.

(R1) A point of P has $X - s = uc - vs \in [-s, bc]$, so every row lies in $0 \leq X \leq W$. *(R2)* The shift $(0, \sec)$ changes v by exactly 1, so in row 0's frame row j lies in $\{j \leq v \leq j+1\}$ and the rows have disjoint interiors. *(R3)* The lowest point of row j is its corner, at height $y_0 + j\sec$, and its highest is at height $y_0 + j\sec + h$. So the rows fit in $[0, L]$ when $y_0 + (R-1)\sec + h \leq L$. Let $R = R(L) = \lfloor (L-h)/\sec \rfloor + 1$, the most rows with $y_0 = 0$, and let $F = L - (R-1)\sec - h \in [0, \sec)$ be the free length.

End regions. With row 0 at lift y, let $\ell(X) = y + (X-s)T$ for $X \geq s$ and $\ell(X) = y + (s-X)/T$ for $X \leq s$. *(E1)* Every row lies in $\{Y \geq \ell(X)\}$: in row 0's frame a point of any row has $u, v \geq 0$, and $Y - y - (X-s)T = vc + vs^2/c \geq 0$ for $X \geq s$, while $Y - y - (s-X)/T = us + uc^2/s \geq 0$ for $X \leq s$. So a unit square in $\{0 \leq X \leq W,\ 0 \leq Y \leq \ell(X)\}$ meets no row in an interior point.

Fill. Let $a_i(y) = \max(0,\ s + (i+1-y)\tau)$ and $m_i(y) = \max(0, \lfloor W - a_i(y) \rfloor)$. Level i holds the m_i unit squares $[W - m_i + q,\ W - m_i + q + 1] \times [i, i+1]$, $0 \leq q < m_i$, and $\mathrm{fill}(y) = \sum_i m_i(y)$. *(E2)* These squares lie under ℓ. If $a_i \geq s$, ℓ increases on $[a_i, W]$ and $\ell(a_i) \geq i+1$. If $a_i < s$, then $i + 1 < y \leq \min \ell$.

(E3) The 180° rotation about $(W/2, L/2)$ maps the rows onto rows with lifts $y_1 + (R-1-j)\sec$, where $y_1 = L - y_0 - (R-1)\sec - h$. So the region over the last row is a rotated end region with lift y_1, and takes the rotated fill. The two fills are disjoint, since each vertical line $X = \text{const}$ meets row 0 and row $R-1$. A band of length L therefore holds

$$C(L) = bR + \mathrm{fill}(y_0) + \mathrm{fill}(y_1)$$

unit squares. Below, $y_0 = 0$ and $y_1 = F$.

3. The packing

Let $S = k - b + W < k$, and suppose $k - b > h$. Put three pieces in $[0, S]^2$:

- the aligned grid $[0, k-b]^2$;
- the band of length S in $[k-b, S] \times [0, S]$ (with $x = k - b + X$, $y = Y$);
- the band of length $k-b$, reflected in $y = x$, in $[0, k-b] \times [k-b, S]$.

They have disjoint interiors. Since $k^2 - (k-b)^2 = bk + b(k-b)$,

$$w(k) \leq b(k-S) + \big[bS - C(S)\big] + \big[b(k-b) - C(k-b)\big]. \tag{1}$$

4. The end lemma

From now on b is odd. Put $\eta = b/2 - \tau$; at t_0, $\eta_0 = 1/(2b)$. Say $t > 1/b$ is *close* if

(a) $\lfloor W - s - \tau \rfloor = (b-1)/2$;
(b) $0 \leq \eta < 1/2$;
(c) $2\tau - bc \geq 0$;
(d) $h < 3$.

At t_0 we have $W_0 - s_0 - \tau_0 = (b-1)/2 + \delta_b$ with $\delta_b = \frac{1}{2} + \frac{1}{2b} - \frac{2b}{b^2+1} \in (0,1)$; here $\delta_b > 0$ because $b^3 - 3b^2 + b + 1 > 0$ for $b \geq 3$. Also $2\tau_0 - bc_0 = (b^2-1)/(b(b^2+1)) > 0$. So (a)–(d) hold strictly at t_0, and by continuity on an interval $(1/b, 1/b + \varepsilon_b)$.

Lemma 2. *If t is close, then* $\mathrm{fill}(0) \geq (b-1)/2$, *and for* $0 \leq y < \sec$,

$$\mathrm{fill}(y) > \tfrac{b-1}{2} + 2z - 1 - 2\eta, \qquad z = W - s - 2\tau + y\tau.$$

Proof. $\mathrm{fill}(0) \geq m_0(0) = \lfloor W - s - \tau \rfloor = (b-1)/2$ by (a). For $0 \leq y < \sec$ and $i \geq 0$, the clipping in a_i is inactive:

$$s + (i + 1 - y)\tau > s + (1 - \sec)\tau = c(1-c)/s \geq 0.$$

So level 0 holds at least $\lfloor z + \tau \rfloor$ squares and level 1 at least $\lfloor z \rfloor$. Since $\tau = \frac{b-1}{2} + \frac{1}{2} - \eta$ with $\frac{b-1}{2} \in \mathbb{Z}$, together they hold at least $\frac{b-1}{2} + \lfloor z + \frac{1}{2} - \eta \rfloor + \lfloor z \rfloor$. For $0 \leq \eta < \frac{1}{2}$ and real z with fractional part f, the sum $\lfloor z \rfloor + \lfloor z + \frac{1}{2} - \eta \rfloor$ is $2z - 2f > 2z - 1 - 2\eta$ if $f < \frac{1}{2} + \eta$, and $2z - 2f + 1 > 2z - 1$ otherwise. □

The odd width matters: it puts the arguments of the two floors half a unit apart. For even b the arguments differ by an integer minus $1/(2b)$, and this pairing gives nothing.

Lemma 3. *If t is close and $L > h$, then with $R = R(L)$, $y_0 = 0$ and $y_1 = F$,*

$$bL - C(L) < bL(1-c) + bch - 2b + 2 - 2(W - s - 2\tau) + 2\eta. \tag{2}$$

At t_0 the right side equals $2bL/(b^2+1) + b + 2 - \kappa_b$, where $\kappa_b = (7b^4+1)/\big(b(b^2+1)^2\big)$.

Proof. $bR = b + bc(L - h - F)$. By Lemma 2, $C(L) > bR + (b-1) + 2(W - s - 2\tau) + 2F\tau - 1 - 2\eta$. The terms in F sum to $F(2\tau - bc) \geq 0$ by (c), so drop them and rearrange. The value at t_0 uses $1 - c_0 = 2/(b^2+1)$, $bc_0h_0 = 3b - b(10b^2+2)/(b^2+1)^2$, $W_0 - s_0 - 2\tau_0 = -(b^2-1)/(b(b^2+1))$ and $2\eta_0 = 1/b$. □

5. Proof of Theorem 1

Let $k \geq 6$ and let b be an odd integer with $|b - \sqrt{2k}| \leq 1$. Then $b \geq 3$. For $k \geq 8$, $k - b \geq k - \sqrt{2k} - 1 \geq 3 > h$ by (d). For $k = 6, 7$ the only choice is $b = 3$, and $k - b \geq 3 > h$. For close t, (1) and (2) with $L = S$ and $L = k - b$ give $k^2 - N(t) < H(t)$, where $N(t)$ is the number of squares packed and $H(t)$ is $b(k - S)$ plus the two right sides of (2).

This holds for each t separately, even though R, F and the floors jump with t. H is continuous, and as $t \to 1/b^+$, $S \to k$ and

$$H(t) \to B(k) = \frac{4kb}{b^2+1} + 2b + 2 + \frac{2}{b^2+1} - 2\kappa_b.$$

Now $4kb/(b^2+1) < 4k/b$, and $4k/b + 2b = 4\sqrt{2k} + \frac{2}{b}(b - \sqrt{2k})^2 \leq 4\sqrt{2k} + \frac{2}{b}$. So

$$B(k) < 4\sqrt{2k} + 2 - 2\Big(\kappa_b - \frac{1}{b} - \frac{1}{b^2+1}\Big),$$

and $b(b^2+1)^2(\kappa_b - \frac{1}{b} - \frac{1}{b^2+1}) = 6b^4 - b^3 - 2b^2 - b > 0$. Hence $B(k) < 4\sqrt{2k} + 2$. Some close t has $H(t) < 4\sqrt{2k} + 2$, and it gives a packing with $S < k$ and $k^2 - N(t) < 4\sqrt{2k} + 2$. □

Remark (Sharpness). For odd $b \geq 5$ and $k = (b^2 - b + 8)/2$, the construction at close t has waste exactly $4b$, which is $4\sqrt{2k} + 2 - 31/(2b) + O(b^{-2})$; for example $k = 59$, $b = 11$, $w = 44$. So the additive constant 2 cannot be lowered for this construction.

6. Checks and certificates

The construction is implemented in exact rational arithmetic (`lband.py` in [Fan26b]).

- *Admissibility.* For each k the program chooses the squeeze $k - S$ as $1/(10^6 k^3)$, divided by 10 until (a)–(d) and $H(t) < 4\sqrt{2k} + 2$ hold. It decides those conditions exactly, so every layout it returns satisfies the theorem's bound by Lemma 3.
- *A fixed squeeze is not enough.* With a fixed squeeze of 10^{-7} the same recipe has waste $1178 = 4\sqrt{2k} + 2$ at $k = 43218$, which violates the strict bound.
- *Every step checked.* Every inequality in the proof was checked exactly for $6 \le k \le 20000$, and the bound for $6 \le k \le 60000$. The least slack $B(k) - w$ is $8019401/78023840599$, at $k = 19705$.
- *Certificates.* Strict certificates (every wall and pair margin positive) of the construction for $6 \le k \le 40$ and $k = 101, 300, 600$ pass our checker and Evan Daniel's [Dan26]. The largest has 359860 squares. A program that is not all-pairs runs both checkers unmodified on overlapping 4×4 windows: two unit squares whose centres are more than $\sqrt{2}$ apart cannot overlap, and both checkers accept such pairs without a separating-axis test.

The construction is not optimized. Searching over b, the number of rows and the split of the free length between the two ends gives waste at most $4\sqrt{2k} + 1.18$ for $15 \le k \le 300$; for example 9946 squares below side 100. A bounded search for better end and corner pieces by local optimization found gains of one or two squares only (at $k = 25, 30, 40$), not growing with b. That is consistent with the heuristic that a boundary between the aligned and the tilted lattice wastes about $1/2$ per unit length.

7. Comparison

The integer form of the theorem is $c^*(k) \le \lceil \sqrt{32k} \rceil$.

- *Ryu's Theorem 1.4.* Ours is smaller at $k = 6$ and for every $k \ge 9$; they tie at $k = 7$, and Ryu's is smaller at $k = 8$.
- *Ryu's Theorem 1.3*, optimized over b. Ours is smaller for $k \ge 14$.
- [Fan26a]. Ours is smaller from $k = 42$ (tie at 40, 41), and larger for $15 \le k \le 39$.
- *Superseded* by Ryu's Theorem 1.2 for $k \ge 1.6 \cdot 10^7$, by his Theorem 1.1 tiers beyond that, and by his computer-assisted values: at $k = 10^5, 10^6, 10^7$ they give $c^* \le 1583, 3972, 10039$, against our $1789, 5657, 17889$.

Within $42 \le k < 1.6 \cdot 10^7$, apart from those five tabulated values, Theorem 1 is the smallest explicit bound we know of.

Acknowledgements and credit

The L-shaped packing, its end-region framework and the bounds compared above are Sungjoon Ryu's. The certificate format and one of the two checkers are Evan Daniel's. The Squares Project (Joshua Levy) keeps the register in which these bounds are compared. The construction code, checkers and certificates were written and run with Claude (Anthropic) as a coding and research agent, directed and reviewed by the author. Two blind language-model reviews of the proof (Claude Opus; OpenAI Codex, GPT-6 Astra) found errors that were corrected. That is not an external mathematical review.

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